TIANWEN V2 · DOMAIN DISCOVERY REPORT

天文 V2 POS

Business Domain Report

Restaurant Point-of-Sale Management System · 商業模型、運作流程與資料結構全景分析報告

VERSION v2.10.0

PLATFORM PWA · React 19 · SQLite WASM

GENERATED 2026-04-11

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1. Executive Summary

Tianwen is a **Progressive Web App (PWA) restaurant POS system** designed for small-to-medium dining operations in Taiwan. The app enables **offline-first** order management, employee time tracking, and cloud-based backup with real-time analytics.

Dimension	Description
Value Proposition	Offline-capable ordering; Cloudflare R2 cloud backup; multi-device sync; analytics dashboard with daily/product-level KPIs
Operational Modes	Order entry (POS), clock-in / clock-out, attendance records, product & category config, analytics
Tech Foundation	React 19 • SQLite WASM (OPFS) • Zustand • TanStack Query + Router • Vercel Serverless • Cloudflare R2
Persistence	SQLite WASM in OPFS for offline; Vercel Functions as API; Cloudflare R2 for versioned backups
Device Support	iPad 10 (2022) and later; portrait orientation enforced for touch usability
i18n	Traditional Chinese (zh-TW) primary; English (en) secondary
User Roles	Operator (order entry), Manager (employee/product admin), System Admin (backup/settings)
Backup Policy	Manual + automatic (daily/weekly schedule); max 30 backups retained per device; gzip compression
Error Handling	Multi-layer error boundaries, error overlay UI, SQLite-persisted error logs, user-facing recovery prompts
Auth Strategy	Google OAuth for staff management binding only; no global user login required (device-based operation)

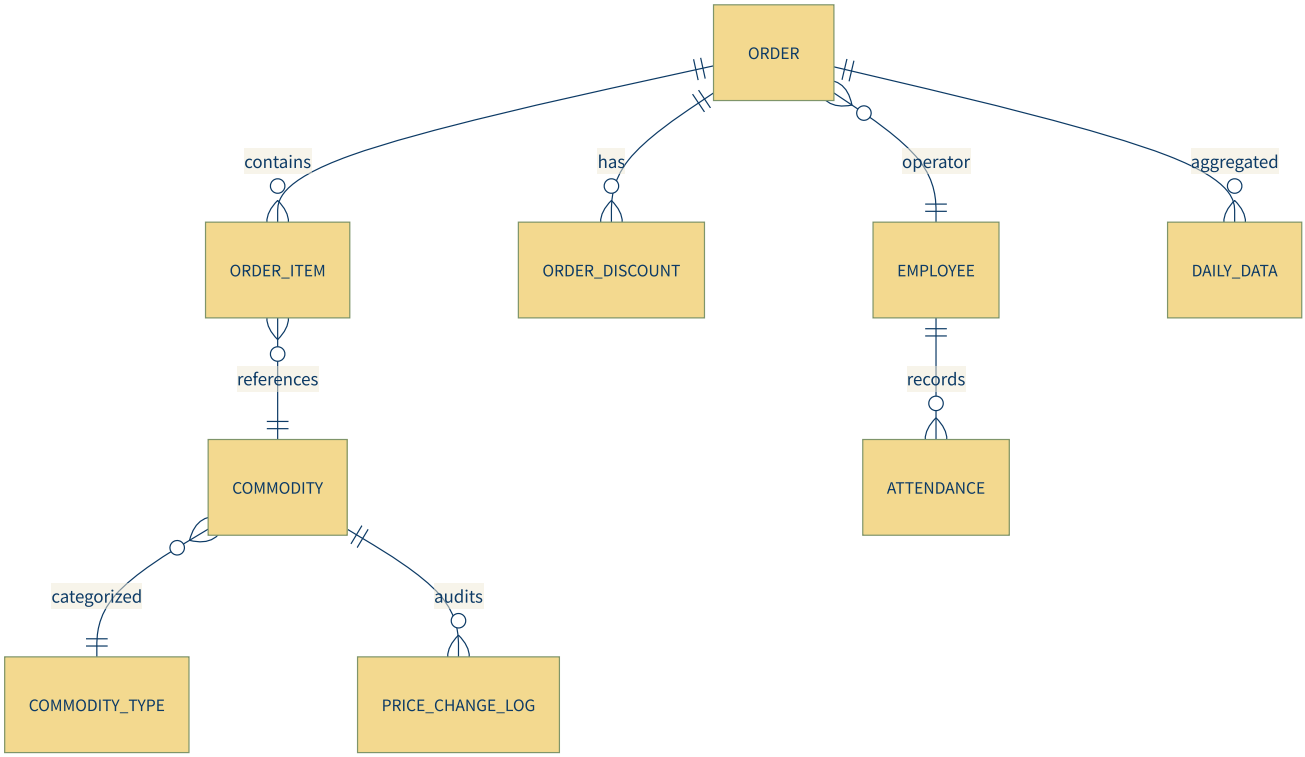
2. Business Entities & Domain Model

The domain is modelled as 13 SQLite tables. The table below summarises each entity's purpose and role; detailed field definitions live in `src/lib/schema.ts` and `src/lib/repositories/*`.

#	Entity	Table	Purpose
2.1	**Order**	`orders`	Core transaction record — single customer order with items, discounts, fulfillment state, operator identity
2.2	**OrderItem**	`order_items`	Normalized line item — single commodity on an order, captures **snapshot price** (immune to later price edits)
2.3	**Commodity**	`commodities`	Menu item definition — product sold by the restaurant (bento, drink, stall item, …)
2.4	**CommodityType**	`commodity_types`	Product category (攤位, 餐盒, 單點, 飲料, 水餃) — groups commodities and drives stall auto-memo rule
2.5	**Employee**	`employees`	Staff record — operator identity for order entry, clock-in, admin privileges, Google OAuth binding
2.6	**Attendance**	`attendances`	Daily time record — clock-in, clock-out, leave/vacation type per employee per day
2.7	**OrderDiscount**	`order_discounts`	Discount line — promotional credit applied to an order (label + amount)
2.8	**BackupLog**	`backup_logs`	Audit trail for all backup operations — success/failure, filename, size, duration, type (manual/auto/v1-import)
2.9	**ErrorLog**	`error_logs`	Runtime error persistence — message, stack, source, timestamp for post-mortem diagnostics
2.10	**PriceChangeLog**	`price_change_logs`	Price history — records every commodity price edit for audit and analytics
2.11	**DailyData**	`daily_data`	Pre-computed daily revenue aggregation — fast analytics queries without scanning full `orders`
2.12	**OrderType**	`order_types`	Order classification — fulfillment mode (stall, delivery, pick-up, …)

#	Entity	Table	Purpose
2.13	**CustomOrderName**	`custom_order_names`	User-defined item names from Calculator overlay for fast re-use on future orders

Core relationships

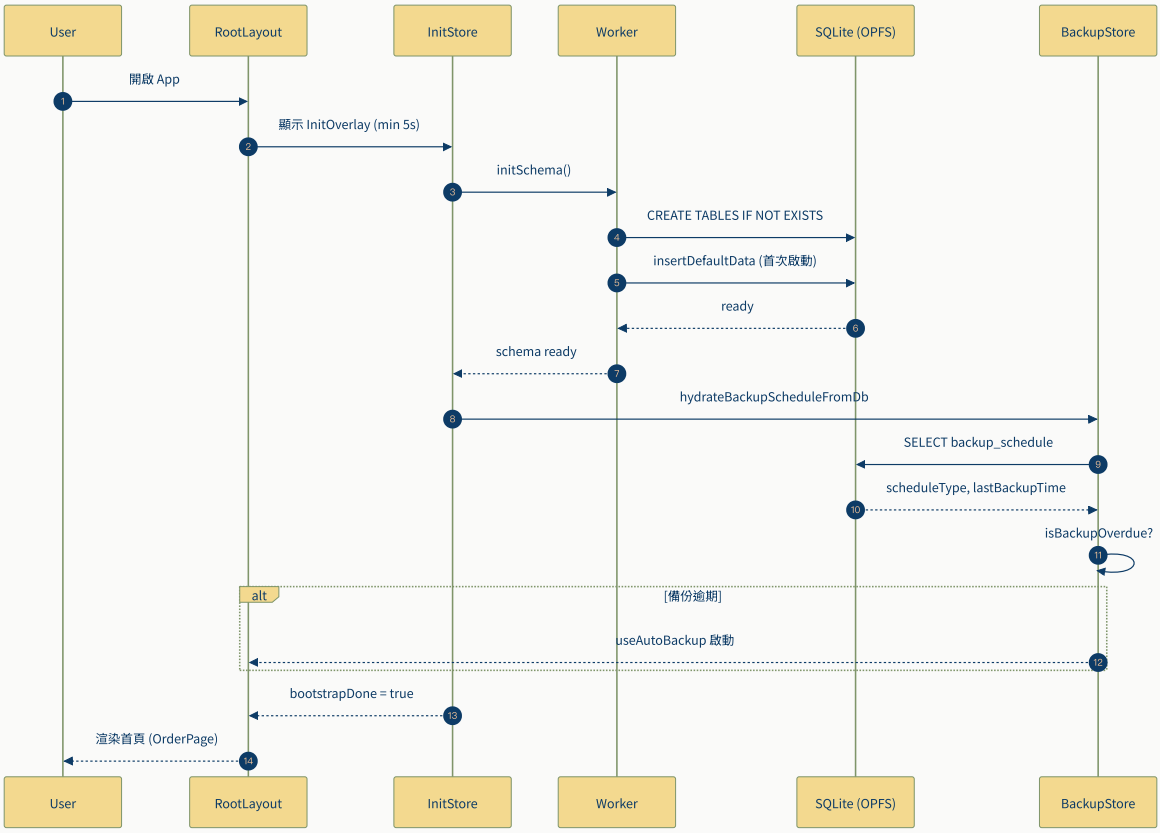


3. Operational Flows

15 end-to-end sequences trace how users and systems interact across the UI, state stores, repositories, and external services. Each flow below gives the entry point, key files, and a sequence or flowchart diagram.

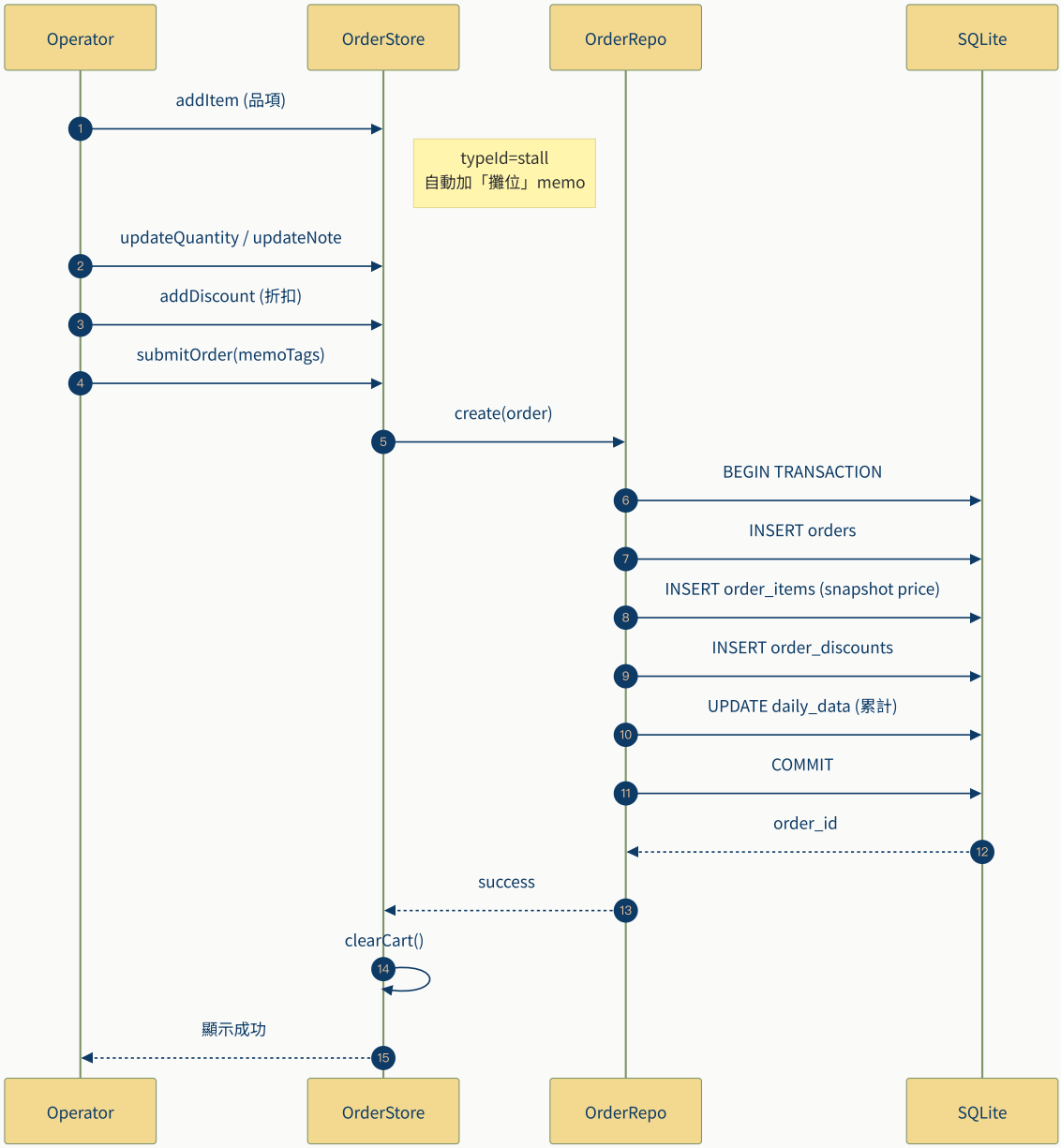
3.1 Bootstrap / Cold Start

Entry: `src/main.tsx` → `src/routes/route-tree.tsx` · **Key:** `useInitStore`, `worker-database.ts`, `schema.ts`, `default-data.ts`



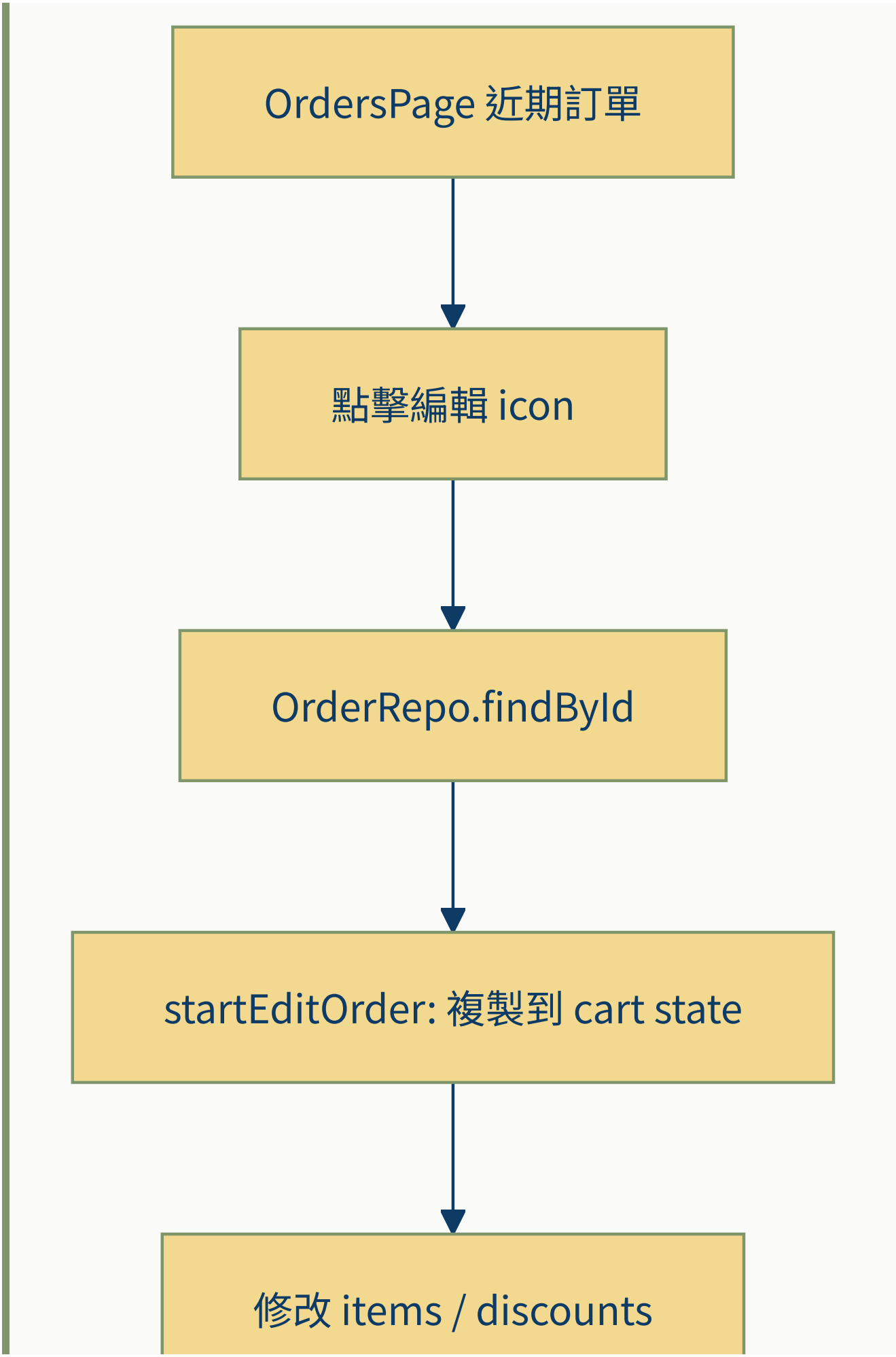
3.2 Order Entry & Submission

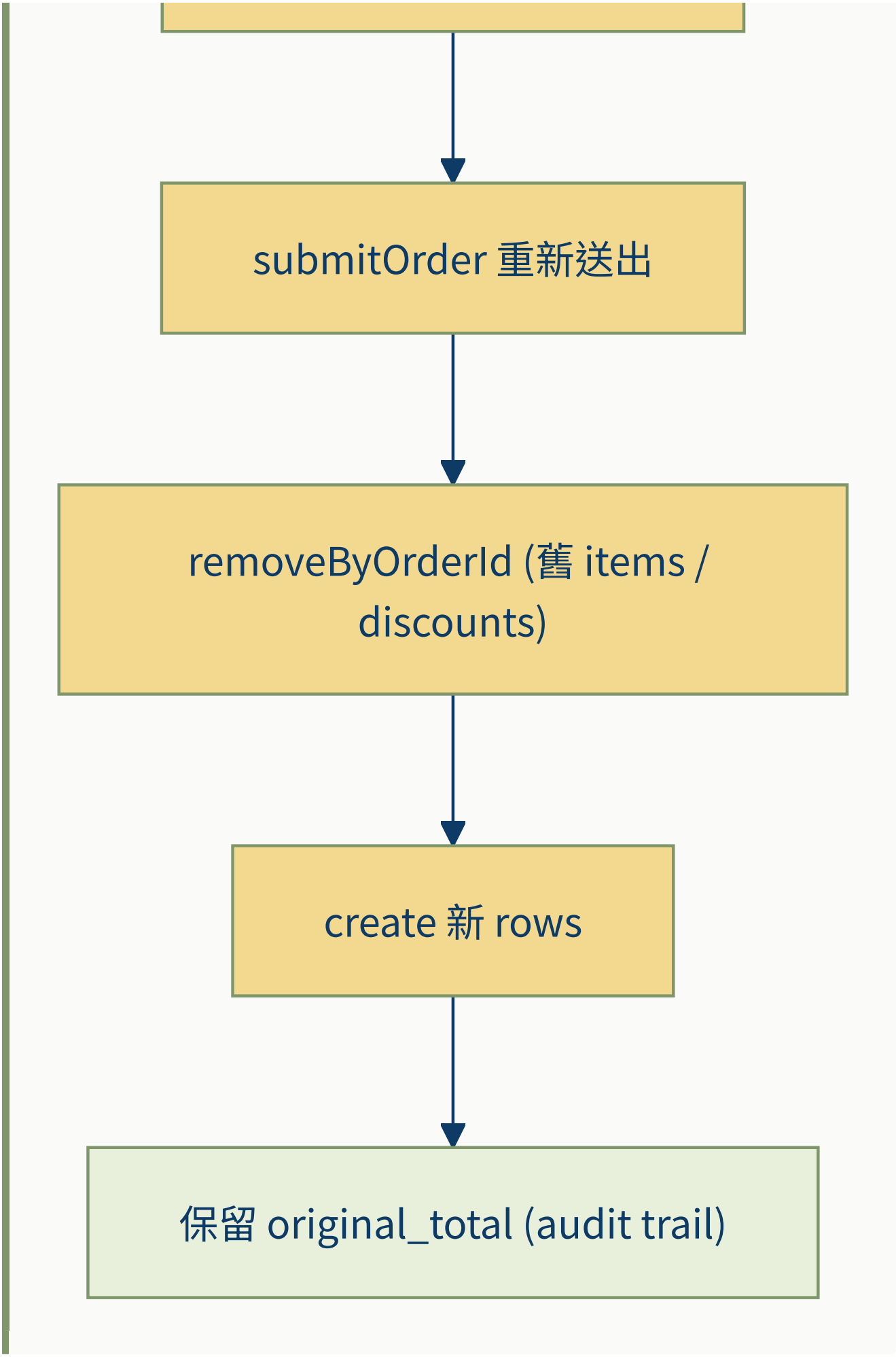
Entry: `src/pages/order` (OrderPage) · **Key:** `order-store.ts`, `order-repository.ts`, `daily-data-repository.ts`



3.3 Order Edit

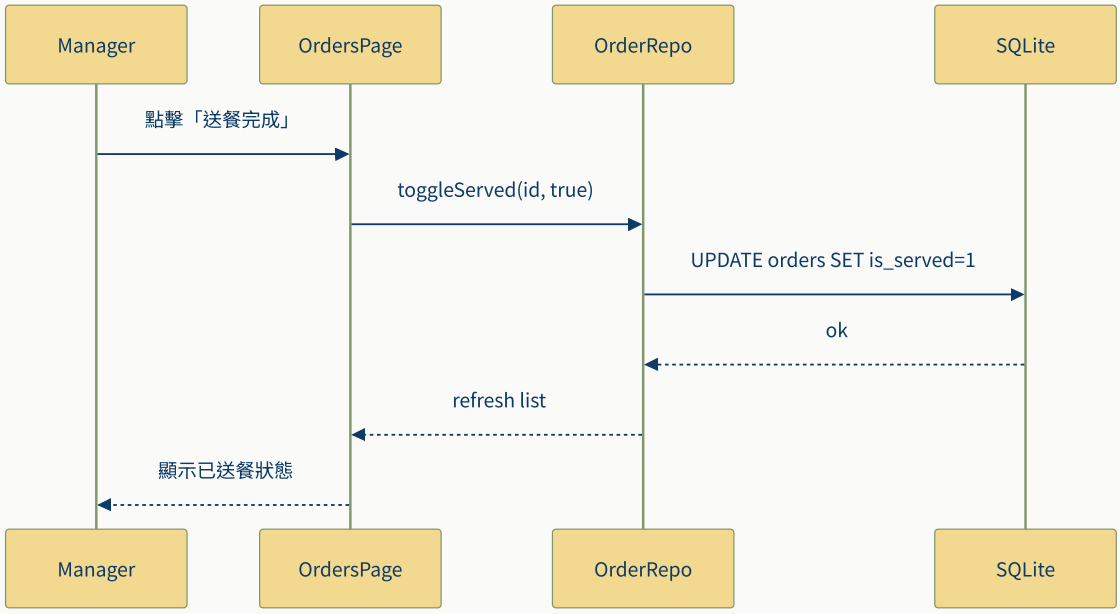
Entry: OrdersPage → 近期訂單 → 編輯 icon · **Key:** `order-store.startEditOrder`, `order-repository.ts`


OrdersPage 近期訂單
點擊編輯 icon
OrderRepo.findById
startEditOrder: 複製到 cart state
修改 items / discounts


submitOrder 重新送出
removeByOrderId (舊 items / discounts)
create 新 rows
保留 original_total (audit trail)

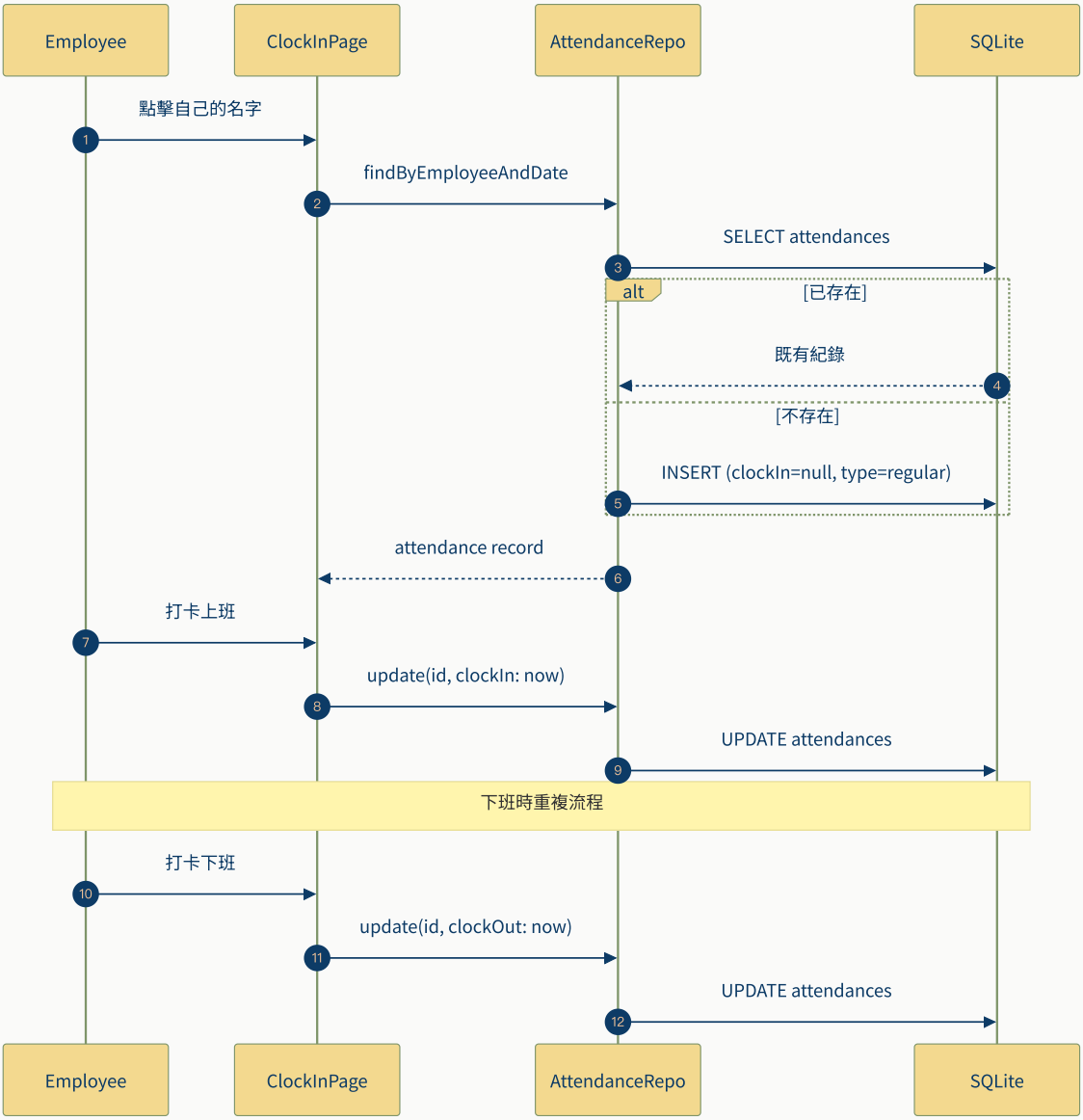
3.4 Order Served Marking

Entry: OrdersPage → 訂單卡片 · **Key:** `OrderRepo.toggleServed`



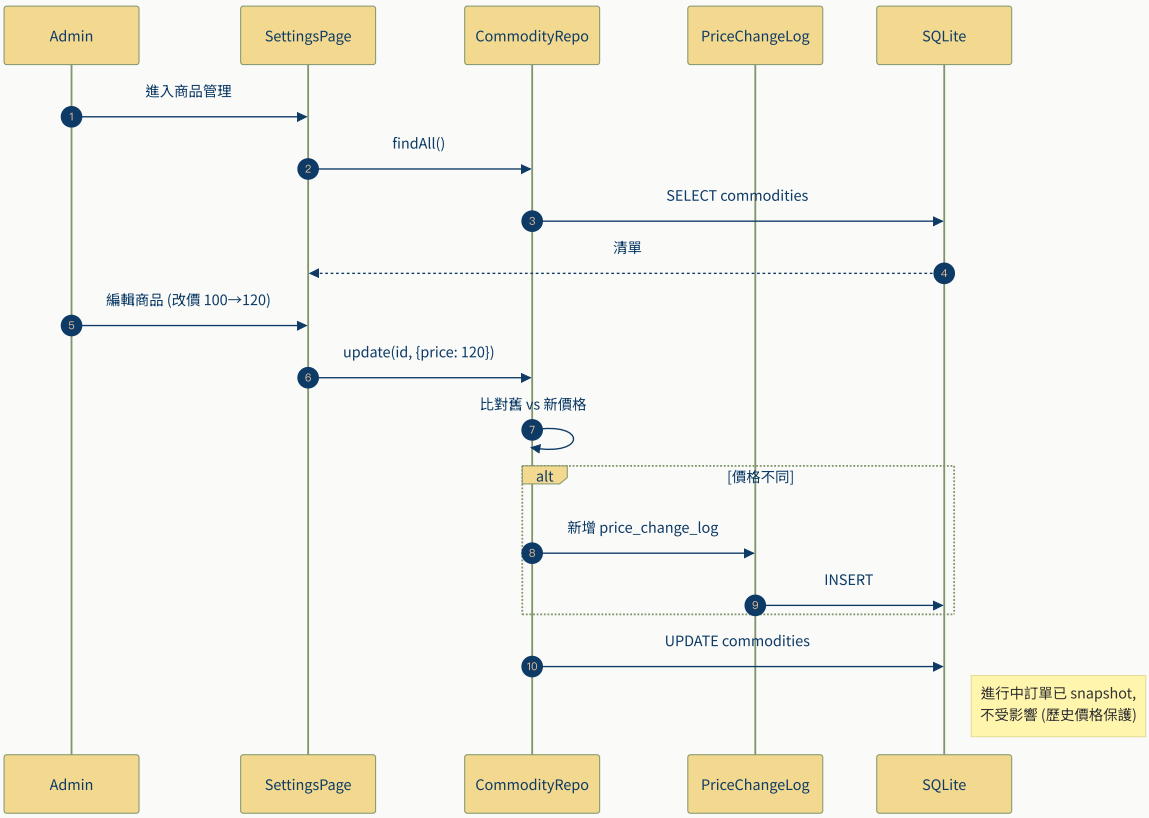
3.5 Clock-In / Clock-Out

Entry: `src/pages/clock-in` (ClockInPage) · **Key:** `attendance-repository.ts`



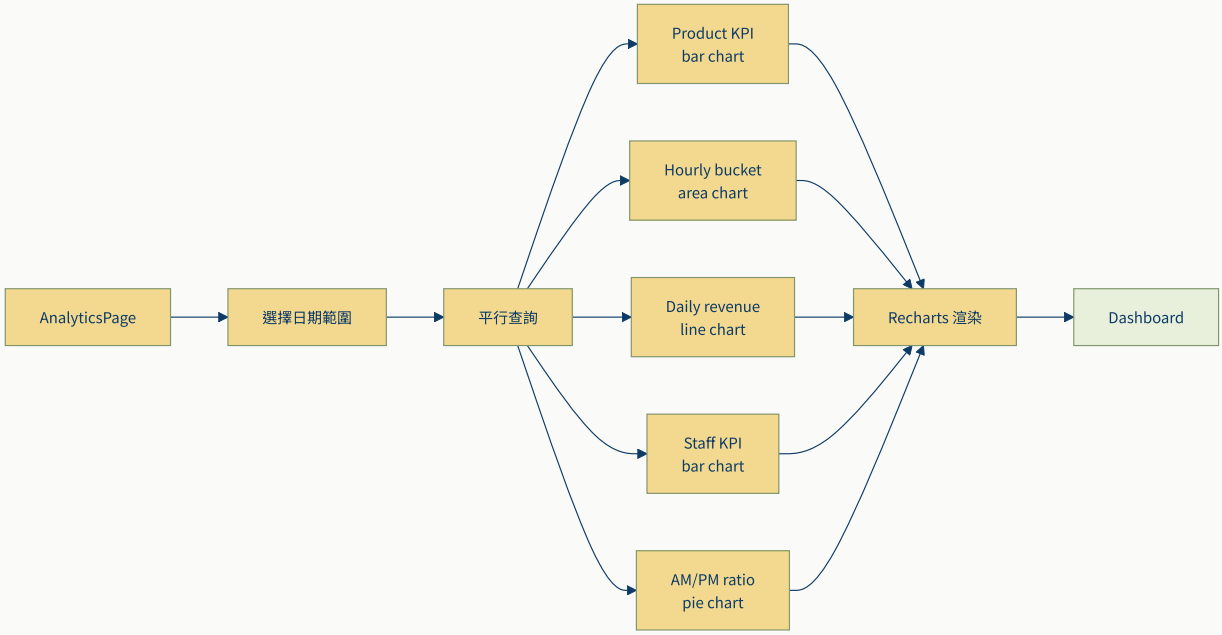
3.6 Product Management & Price Edit

Entry: Settings → 商品管理 · **Key:** `commodity-repository.ts`, `price_change_logs`



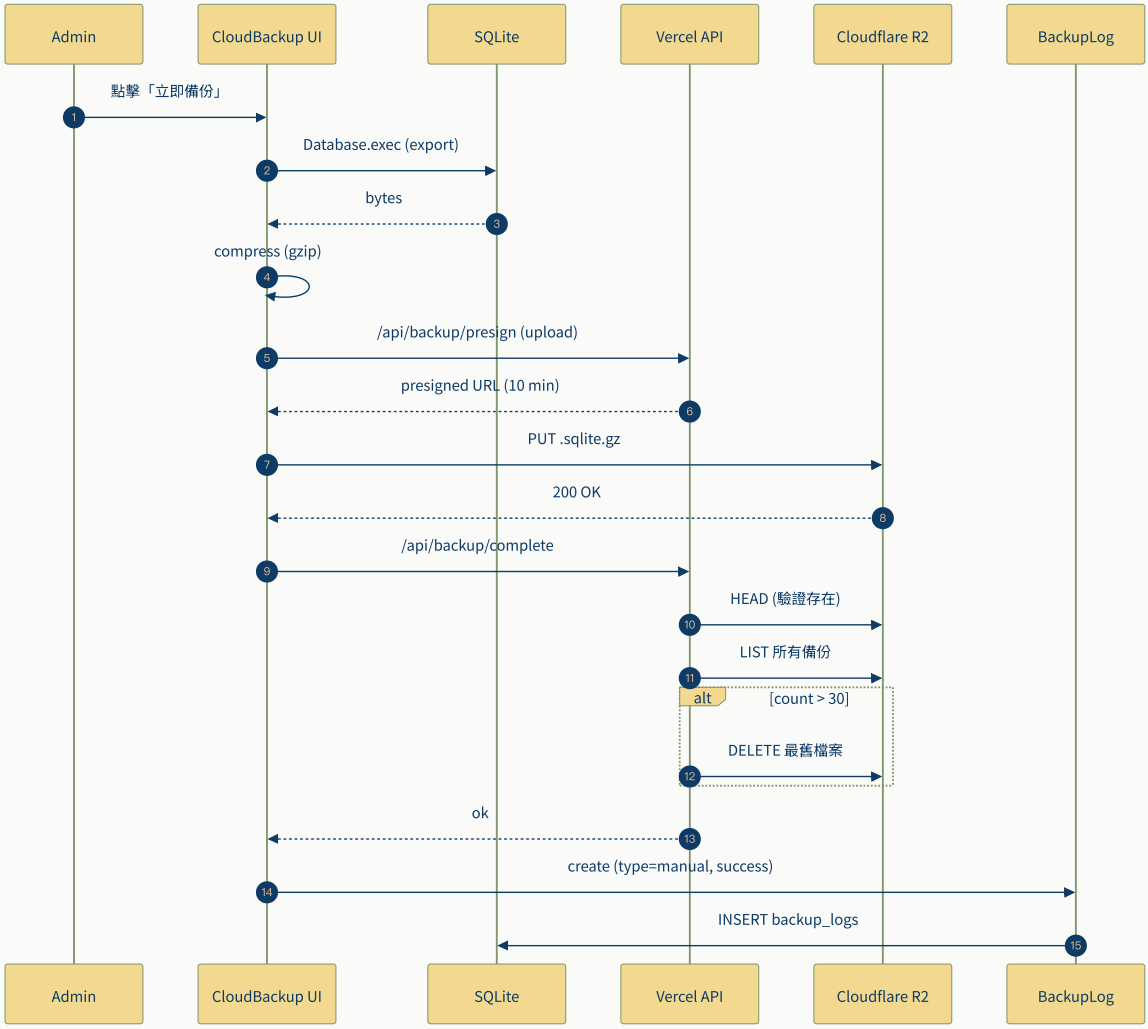
3.7 Analytics Dashboard

Entry: `src/pages/analytics` · **Key:** `statistics-repository.ts`, Recharts



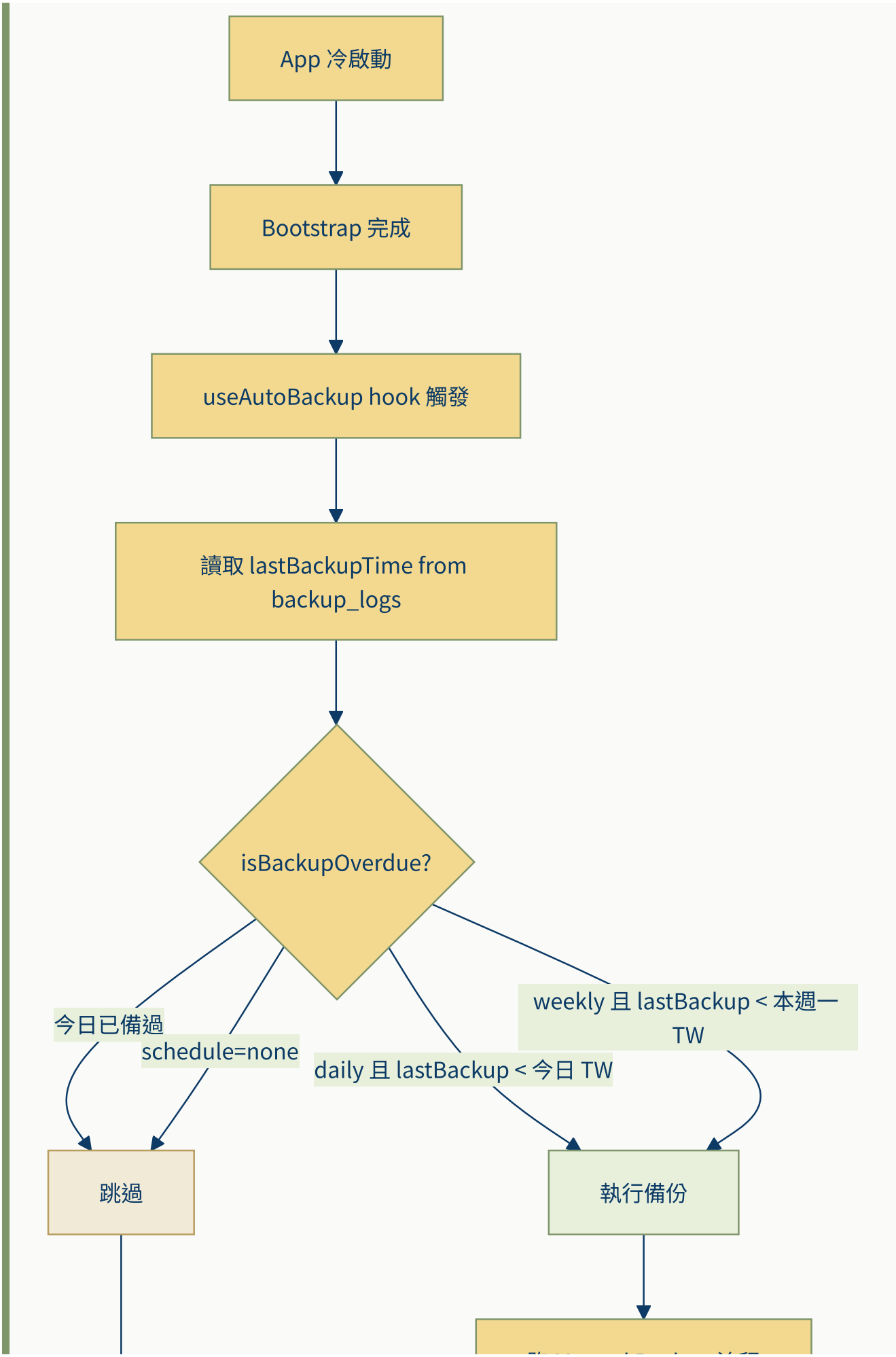
3.8 Manual Backup (to Cloudflare R2)

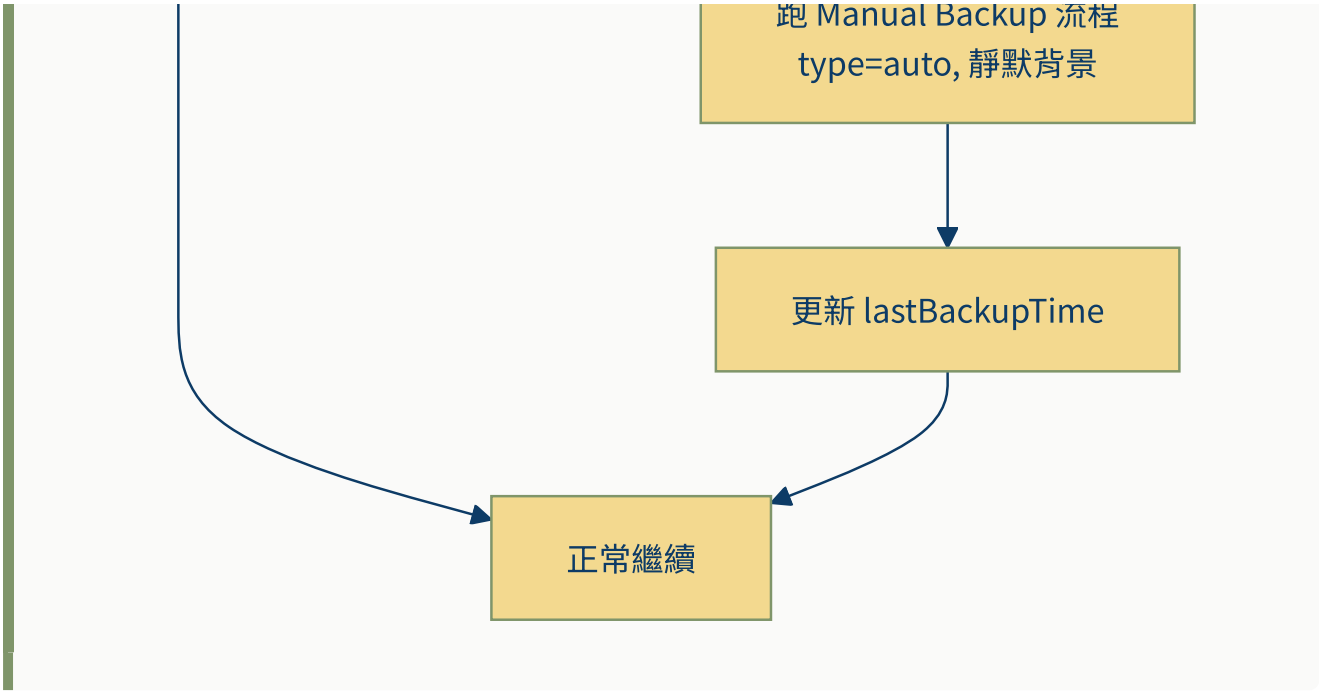
Entry: Settings → 雲端備份 → 立即備份 · **Key:** `backup.ts`, `api/backup/*`, `backup-store.ts`



3.9 Automatic Backup (on startup)

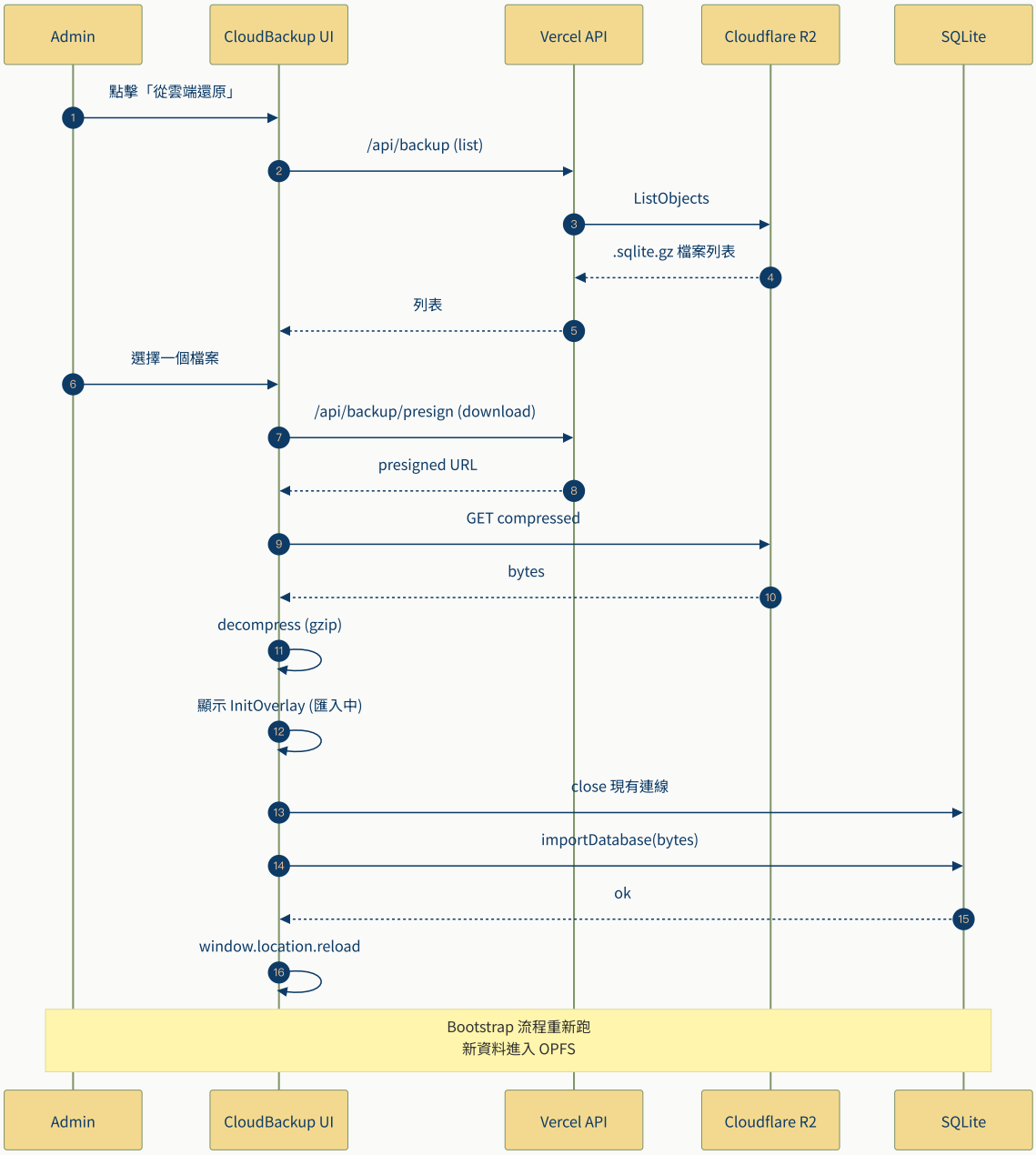
Entry: `useAutoBackup` hook in RootLayout • **Key:** `backup-schedule.ts` (Taiwan timezone aware)


App 冷啟動
Bootstrap 完成
useAutoBackup hook 觸發
讀取 lastBackupTime from backup_logs
isBackupOverdue?
今日已備過
schedule=none
daily 且 lastBackup < 今日 TW
weekly 且 lastBackup < 本週一 TW
跳過
執行備份


跑 Manual Backup 流程
type=auto, 靜默背景
更新 lastBackupTime
正常繼續

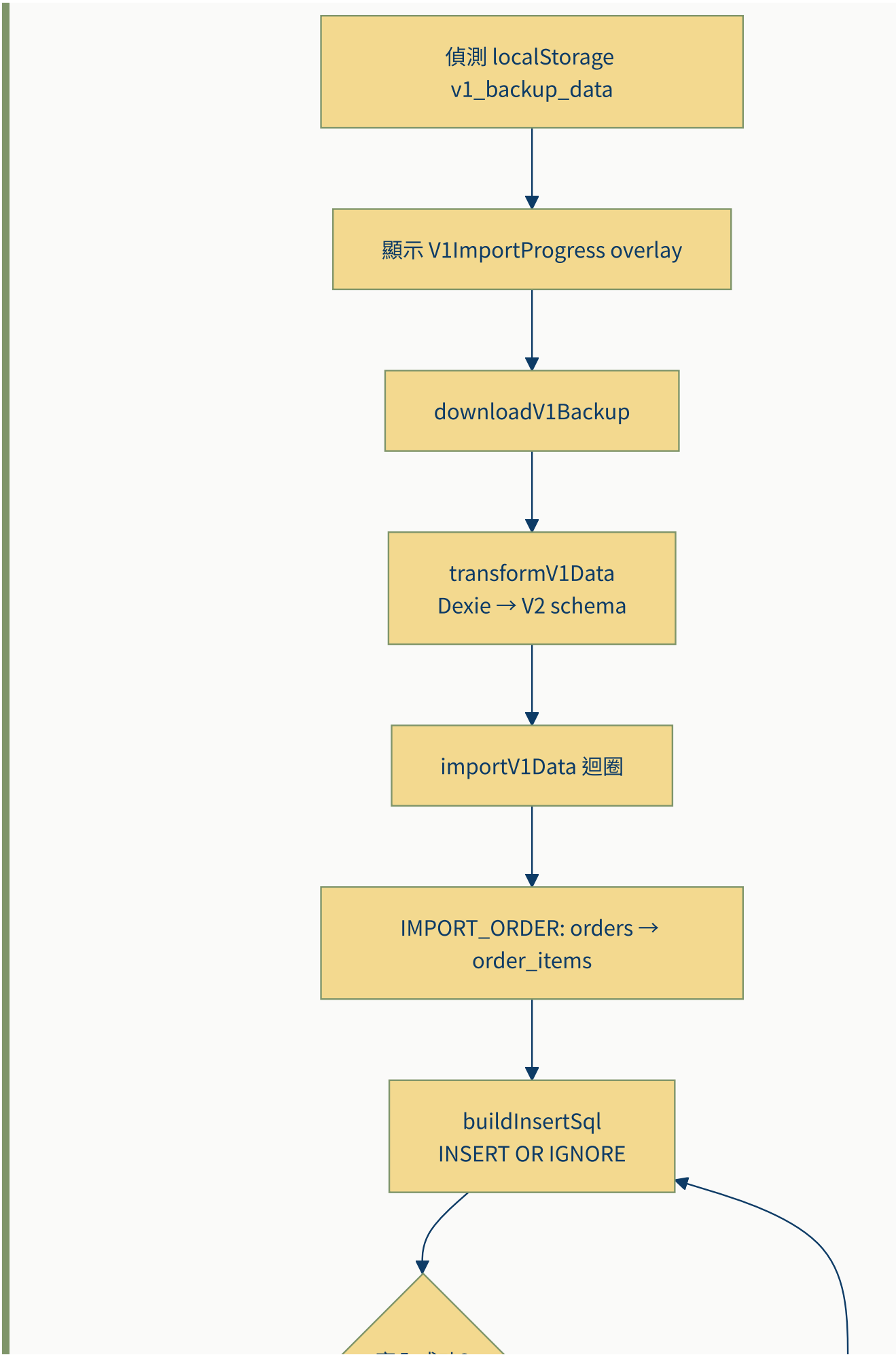
3.10 Cloud Restore (V2 Import from R2)

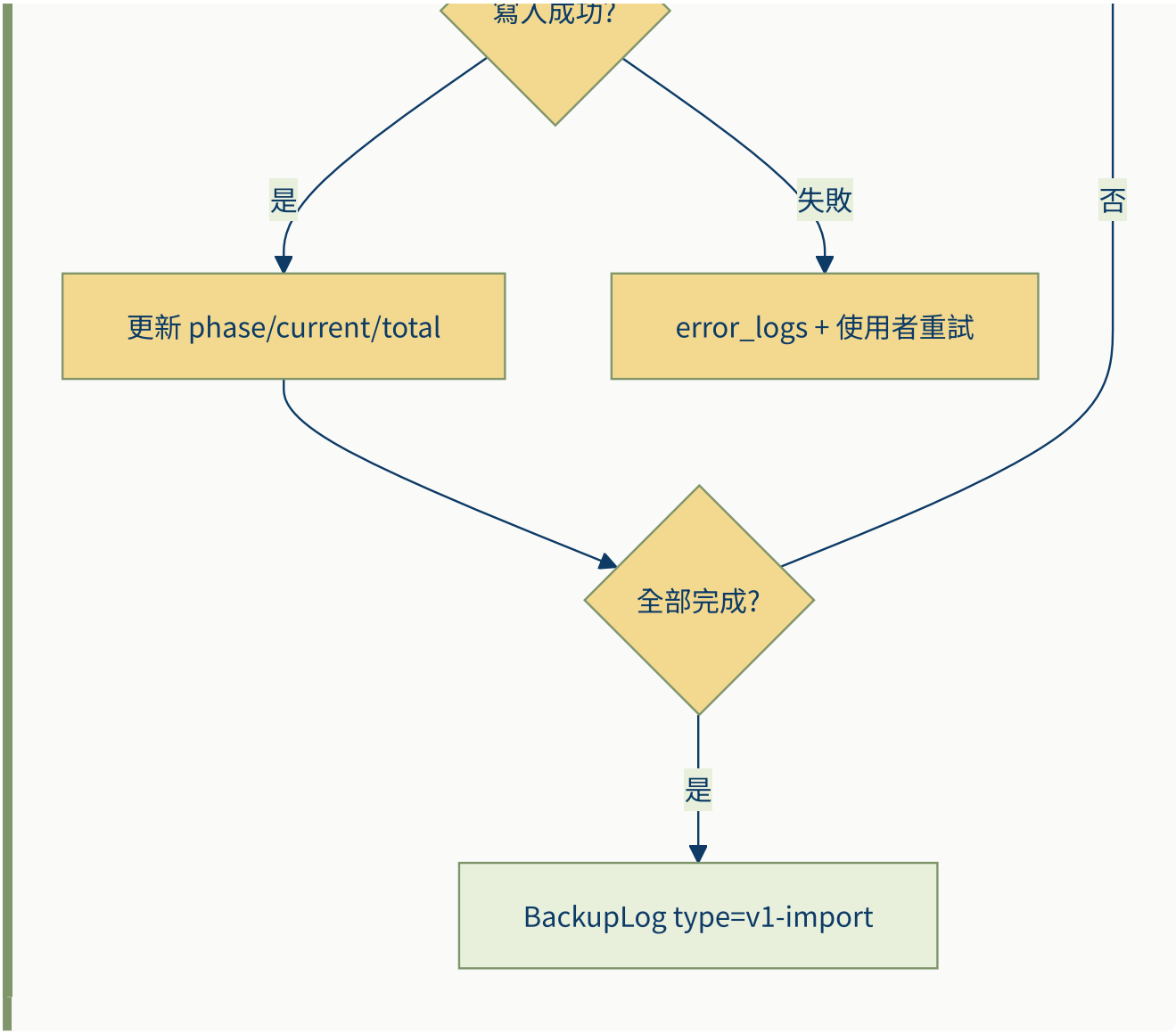
Entry: Settings → 雲端備份 → 還原 · **Key:** `backup.ts::decompress`, `worker-database::importDatabase`



3.11 V1 → V2 Legacy Data Import

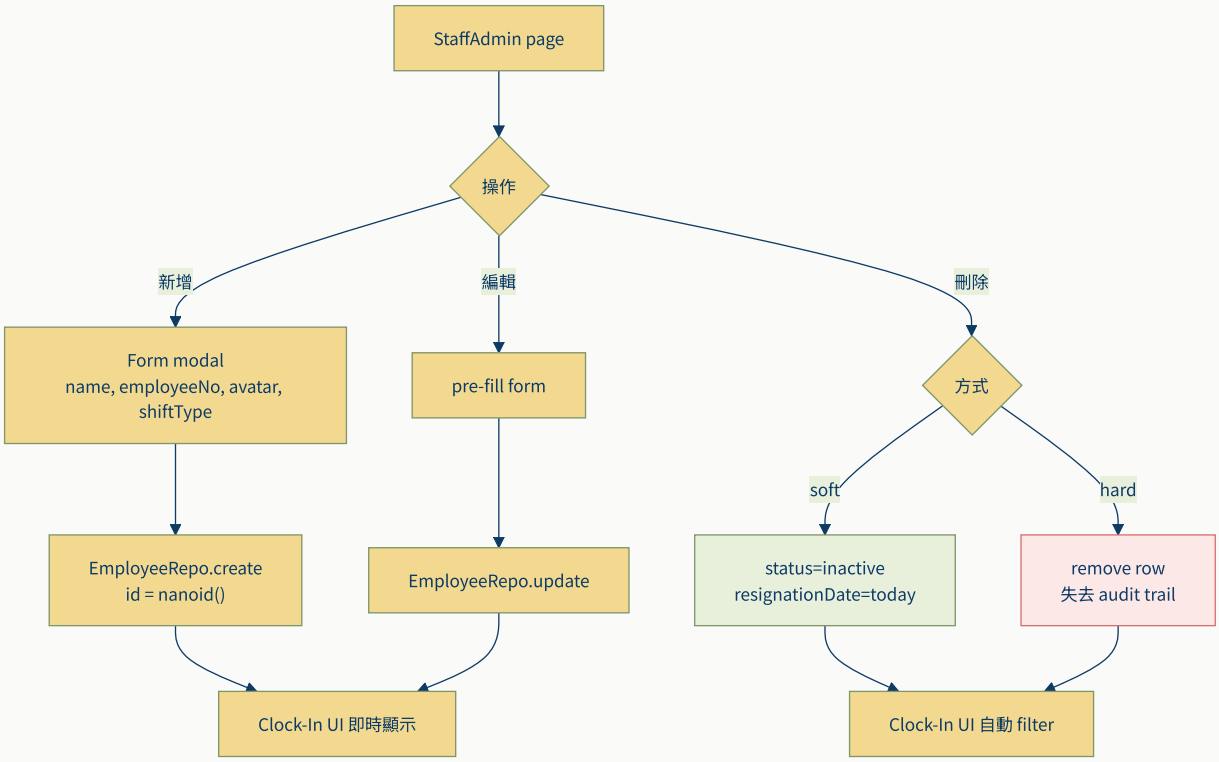
Entry: 首次啟動偵測 V1 backup · **Key:** `v1-data-importer.ts` (INSERT OR IGNORE 保持 idempotent)


偵測 localStorage
v1_backup_data
顯示 V1ImportProgress overlay
downloadV1Backup
transformV1Data
Dexie → V2 schema
importV1Data 迴圈
IMPORT_ORDER: orders →
order_items
buildInsertSql
INSERT OR IGNORE


寫入成功?
是
失敗
否
更新 phase/current/total
error_logs + 使用者重試
全部完成?
是
BackupLog type=v1-import

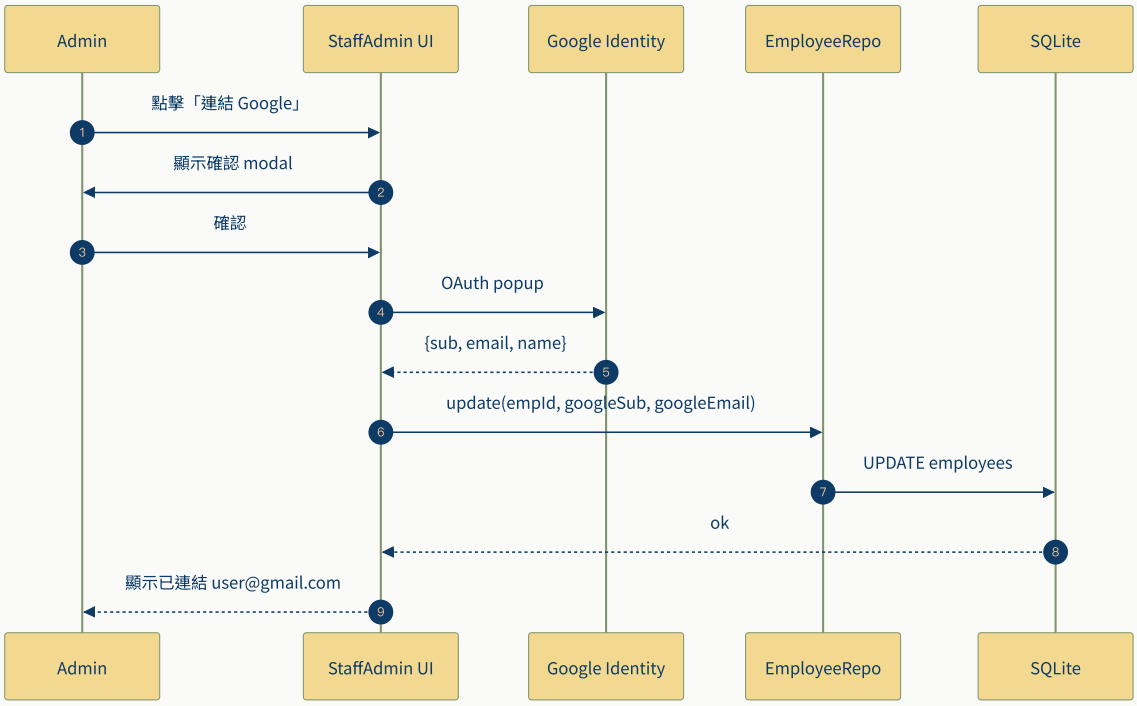
3.12 Staff Management (CRUD)

Entry: Settings → Staff Admin · **Key:** `employee-repository.ts` (soft-delete)



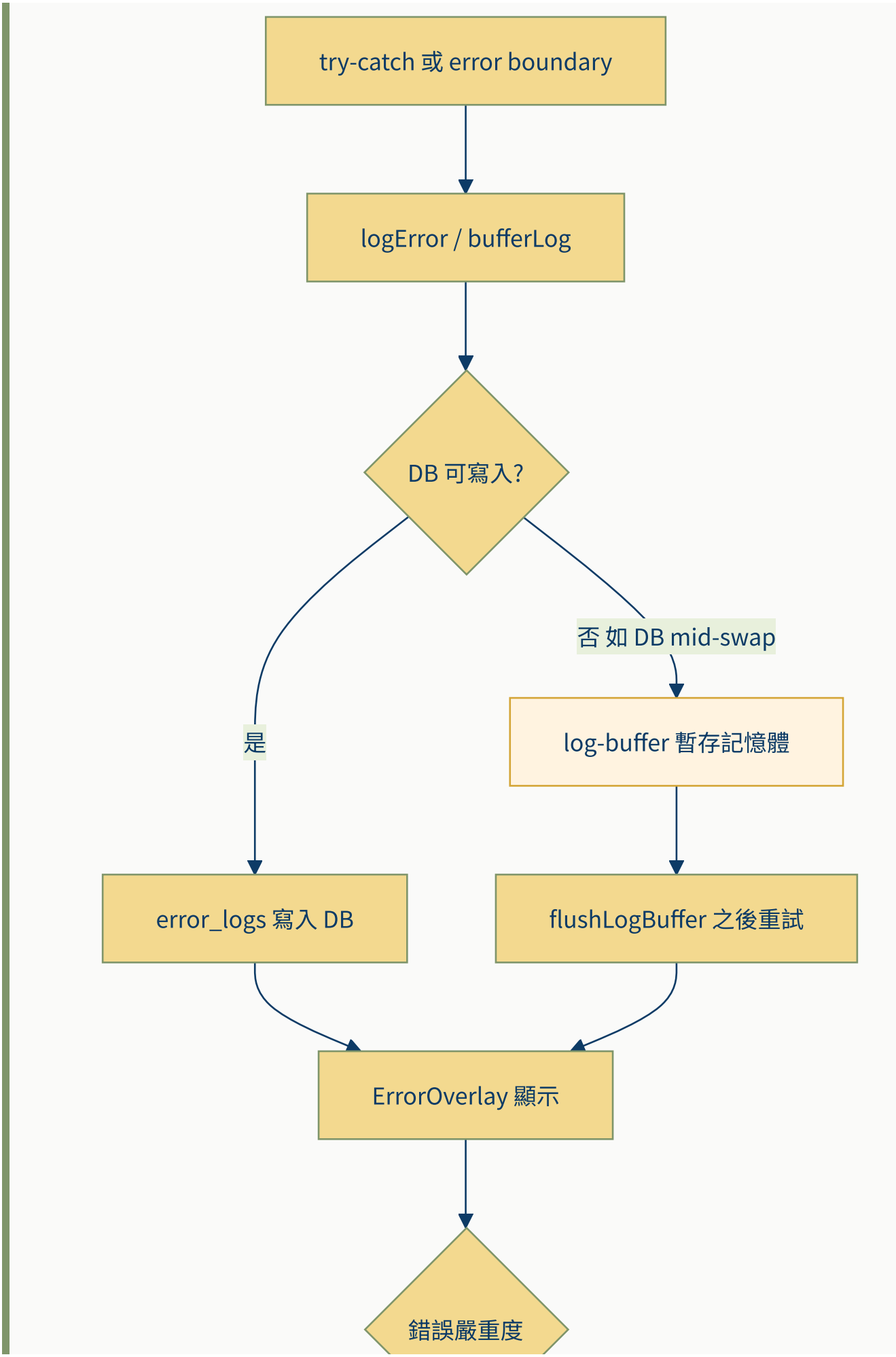
3.13 Google OAuth Binding (per-employee)

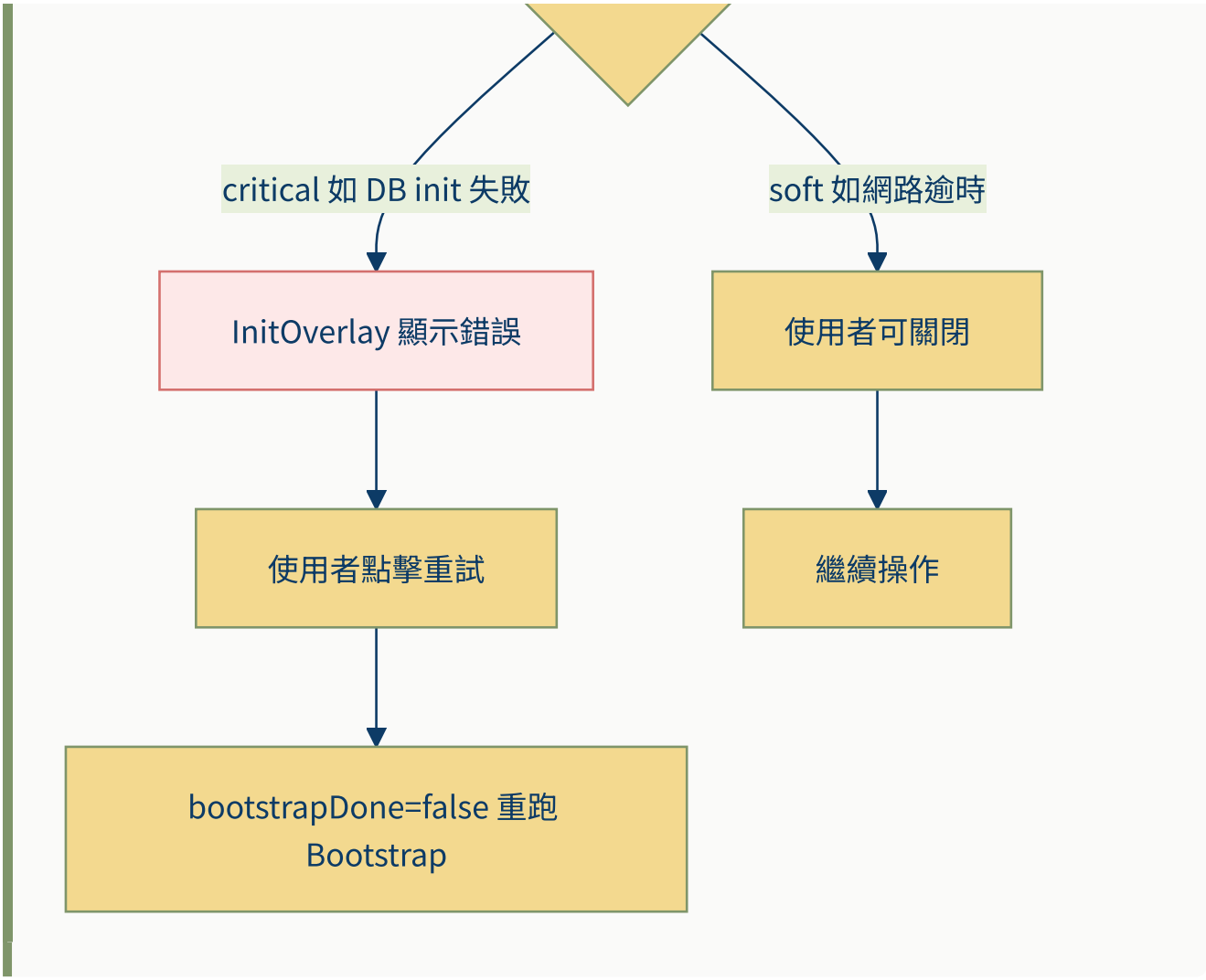
Entry: StaffAdmin → 員工卡片 → 連結 Google · **Key:** Google Identity Services



3.14 Error Handling & Recovery

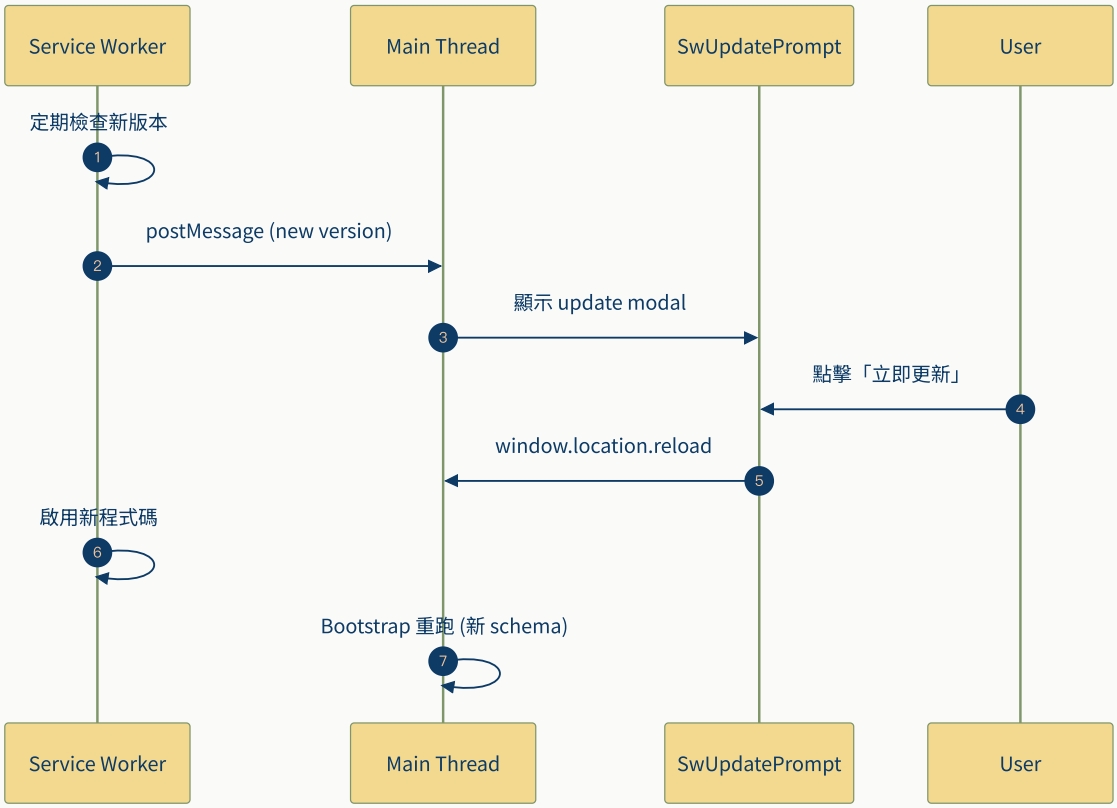
Entry: 任何 try-catch 或 error boundary · **Key:** `error-logger.ts`, `error-ui/`, `log-buffer.ts`


try-catch 或 error boundary
logError / bufferLog
DB 可寫入?
是
否 如 DB mid-swap
log-buffer 暫存記憶體
error_logs 寫入 DB
flushLogBuffer 之後重試
ErrorOverlay 顯示
錯誤嚴重度


critical 如 DB init 失敗
soft 如網路逾時
InitOverlay 顯示錯誤
使用者可關閉
使用者點擊重試
繼續操作
bootstrapDone=false 重跑 Bootstrap

3.15 PWA Update Detection & Installation

Entry: Service Worker 偵測新版本 · **Key:** `sw.ts`, `SwUpdatePrompt`



4. Business Rules & Invariants

The 13 rules below govern how the system behaves beyond the CRUD surface. Each rule lives in a specific place in code; edits that violate these rules should be treated as regressions.

4.1 Stall Category Auto-Memo Tagging

Any order containing commodities with `type_id = "stall"` automatically receives a memo tag 「攤位」. Marks orders as containing side items for kitchen display.

`src/stores/order-store.ts:6–8 · submitOrder(memoTags?) line 82`

4.2 Backup Scheduling with Taiwan Timezone Awareness

Backup overdue checks must respect Taiwan timezone (UTC+8), not server time. `getTaiwanDayStart` and `getTaiwanWeekStart` compute UTC milliseconds. On Tuesday TW 14:00, if last backup was Monday TW 23:00, daily schedule is NOT overdue.

`src/lib/backup-schedule.ts · TAIWAN_OFFSET_MS`

4.3 Backup Retention Policy (max 30 per device)

After successful upload, backend lists all `.sqlite.gz`, sorts by LastModified desc, and deletes oldest if count > 30. Automatic garbage collection ensures storage quota not exceeded.

`api/backup/complete.ts:20 · MAX_BACKUPS = 30`

4.4 Order Snapshot Price Protection

When an order is created, each item's price is captured into `order_items.price`. Subsequent price changes to commodities do NOT affect historical orders — revenue integrity and accounting are preserved.

`src/stores/order-store.ts:63–69 · order-item-repository.ts`

4.5 V1 → V2 Migration Idempotency

Legacy V1 data can be safely imported multiple times without duplication. `INSERT OR IGNORE` ensures interrupted imports can be resumed.

`src/lib/v1-data-importer.ts:78 · buildInsertSql`

4.6 Employee Termination Soft-Delete

Terminated employees are not removed from the DB. `status` set to "inactive" and `resignation_date` recorded. Historical orders/attendance preserved for audit; Clock-In UI filters to active only.

`src/lib/schema.ts:92–93 · CHECK constraint`

4.7 Order Number Sequence (customer-facing)

Each order receives an auto-incrementing `number` field independent of its nanoid `id`, used for receipt printing and customer reference. Computed via `SELECT MAX(number)+1`.

`src/lib/repositories/order-repository.ts · getNextOrderNumber`

4.8 Minimum Init UI Display Duration (5 seconds)

InitOverlay must display for at least 5 seconds even if DB is ready. Prevents jarring flash on fast machines; ensures user perceives app as "doing work".

`src/stores/init-store.ts:66–73 · shownAt timestamp`

4.9 Portrait-Only Orientation Enforcement

App locked to portrait on mobile/tablet. RootLayout detects landscape via `matchMedia('(orientation: portrait)')` and shows WaitingOverlay with rotate prompt.

`src/routes/route-tree.tsx:23–41 · useSyncExternalStore`

4.10 Device Identity Tracking

Each browser/device has a unique DEVICE_ID persisted to localStorage. Backup filenames include device code: `tianwen-<device>-<date>.sqlite.gz` — enables per-device restore and cross-device listing.

`src/lib/device.ts · getDeviceId()`

4.11 No Global User Login Required

App operates without a global user login for order entry. OrderPage loads directly — no operator selection step. Admin functions (staff mgmt, products, backup) are gated by `AuthGuard` + Google OAuth. The store has `operatorId` / `operatorName` fields but no UI sets them (planned feature).

`src/components/auth-guard/auth-guard.tsx · AuthGuard`

4.12 Order Edit Rewrites Items (Original Total Preserved)

Editing a submitted order deletes old `order_items` and `order_discounts` and inserts fresh rows. `original_total` is preserved for audit trail if edits change the total.

`src/stores/order-store.ts · startEditOrder + submitOrder`

4.13 Gzip Compression for Backup Transport

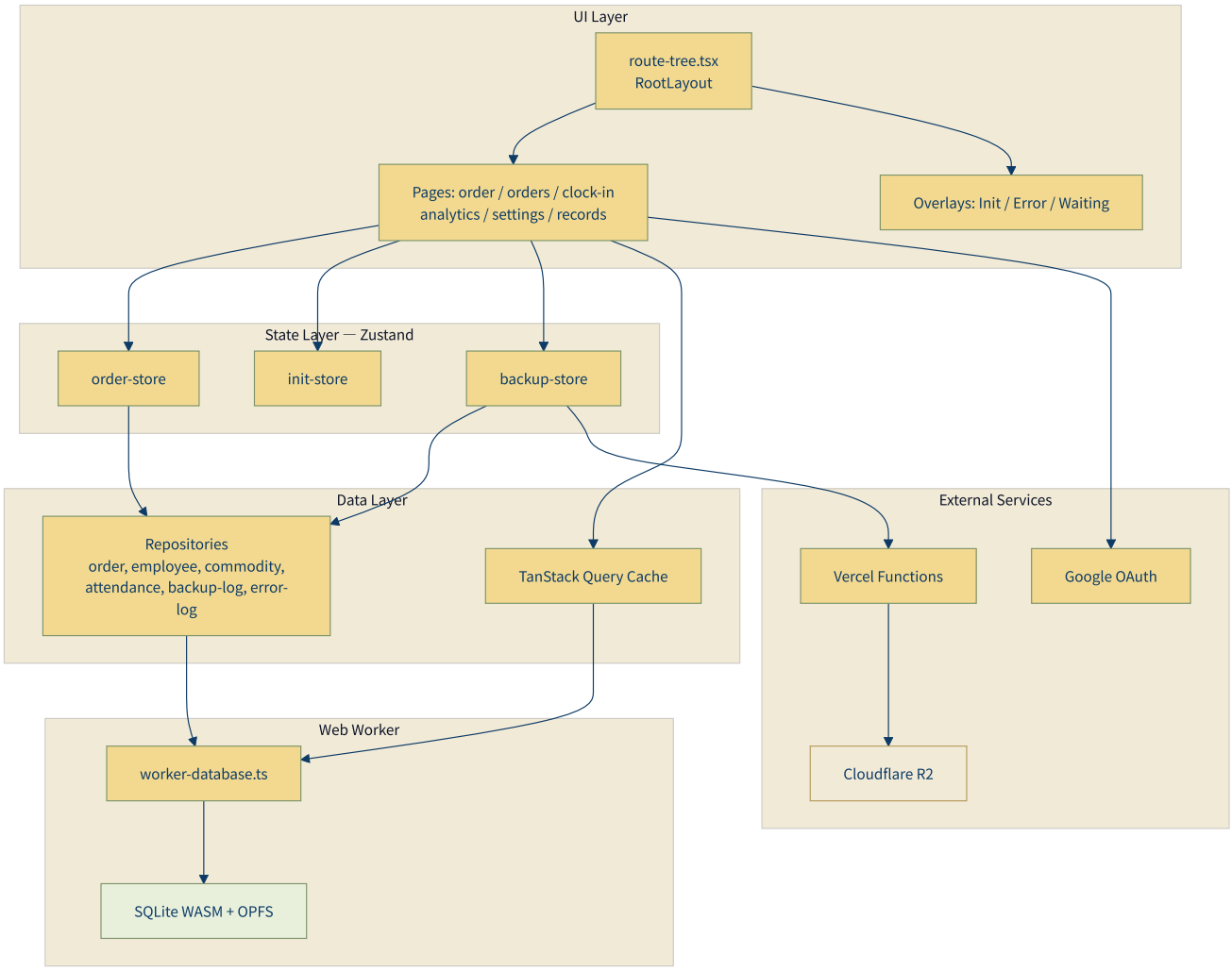
Database backup compressed via browser `CompressionStream('gzip')` before uploading to R2. Typical 5 MB SQLite → 1-2 MB compressed (60-80% reduction).

`src/lib/backup.ts:26–83 · compress / decompress`

5. External Integrations

Integration	Purpose	Notes • Failure Mode
Cloudflare R2	S3-compatible object storage for versioned SQLite backups	Accessed via Vercel Functions with presigned URLs (10-min TTL). Frontend never has R2 credentials. Unavailable → manual export to local still works.
Vercel Functions	Serverless API layer wrapping R2	Endpoints: `/api/backup` (list), `/api/backup/presign` (upload/download URL), `/api/backup/complete` (verify + retention cleanup), `/api/health`.
Google Identity Services	OAuth for per-employee Google account binding	Loaded via `accounts.google.com/gsi/client` script in `index.html`. Not required for order entry; only for admin staff management.
SQLite WASM + OPFS	Browser-local durable storage (opfs-sahpool VFS)	Persistent via Origin Private File System. Path gotcha: SQLite normalises filenames by prepending `/`. All DB access via Web Worker to avoid blocking UI.
TanStack Query	Server-state cache for DB reads (reactive)	Keys drive invalidation on mutations. Used for DB-size queries, cloud backup list, employee list, etc.
Zustand	Client state stores (cart, init, backup, overlays)	`order-store`, `init-store`, `backup-store`. Overlay registration via `useRegisterOverlay` hook with ref-count.

6. Architecture Overview



State boundaries

- **Zustand** — ephemeral client state (cart, active overlays, init progress, backup UI)
- **TanStack Query** — server-state cache over DB reads; invalidation on writes
- **SQLite / OPFS** — durable source of truth; lives in Web Worker to avoid UI blocking
- **localStorage** — DEVICE_ID, v1 backup detection flag, first-run markers

Overlay system

- Three overlays: `InitOverlay` (bootstrap / import), `ErrorOverlay` (error animation), `WaitingOverlay` (portrait rotate prompt)
- Each uses **React portal** to `document.body` to escape ancestor containing blocks from `transform` / `backdrop-filter`

- Registration via `useRegisterOverlay` hook (ref-counted in `init-store.activeOverlays`); AppHeader auto-switches to transparent + white-text styling while any overlay is active

7. Domain Vocabulary (zh-TW ↔ en)

The domain vocabulary is the source of truth for product conversations. Taiwanese restaurant terminology does not always map cleanly to English equivalents.

Category	zh-TW	English Key	Usage
Navigation	點餐	`home`	Order entry page
	訂單	`orders`	Order history
	打卡	`clockIn`	Staff time tracking
	打卡記錄	`records`	Attendance history
	員工管理	`staffAdmin`	Employee management
	統計	`analytics`	Dashboard & KPIs
	設定	`settings`	App settings
Order Entry	攤位	`categoryStall`	Side items (auto-memo tag)
	餐盒	`categoryBento`	Bento boxes
	單點	`categorySingle`	À-la-carte items
	飲料	`categoryDrink`	Beverages
	水餃	`categoryDumpling`	Dumplings
Clock-In	打卡上班	`clockIn`	Start shift
	打卡下班	`clockOut`	End shift
	休假	`applyVacation`	Request leave
	取消休假	`cancelVacation`	Cancel leave request
Staff	員工編號	`employeeNo`	Staff ID
	入職日期	`hireDate`	Start date
	離職日期	`resignationDate`	End date
	班別	`shiftType`	Shift type
	管理員	`admin`	Administrator role
Attendance	一般	`regular`	Normal working day
	特休	`paidLeave`	Paid vacation

Category	zh-TW	English Key	Usage
	病假	`sickLeave`	Sick leave
	事假	`personalLeave`	Personal leave
	缺勤	`absent`	Absent
Backup	裝置代號	`deviceName`	Device identifier
	雲端備份狀態	`cloudBackup`	Cloud backup status

Generated from domain discovery findings • Tianwen V2 POS v2.10.0 • 2026-04-11
Source: `/tmp/tianwen-business-domain-report.md` • Discovery via Claude Code Explore agent